

Adding a Second Retailer

In Week 4 the warehouse supplied one retailer. Now it supplies two. The warehouse does not care which retailer it is sending to - it just needs enough stock to cover both shipments. The retailers are completely independent of each other. Each has its own demand, its own inventory, and its own balance equation.

Warehouse Storage Capacity

Until now the warehouse could hold unlimited stock. That is not realistic. Every physical warehouse has a maximum storage capacity .

$$I_w[t] \leq CAP_STORE \quad \text{for all } t$$

Warehouse cannot hold more than CAP_STORE units at the end of any period

```
# Option 1: upper bound on variable (no shadow price available)
I_w = pulp.LpVariable.dicts('I_w', range(T), lowBound=0, upBound=CAP_STORE)

# Option 2: named constraint (shadow price available)
I_w = pulp.LpVariable.dicts('I_w', range(T), lowBound=0)
for t in range(T):
    m += I_w[t] <= CAP_STORE, f'storage_cap_t{t}'
```

Shadow Prices : What They Are and How to Read Them

A shadow price is the answer to this question: if I relax this constraint by one unit, how much does my total cost change?

1

Shadow price = 0

The constraint is not active. The warehouse never actually reaches its capacity limit. Adding more capacity would not change anything because you are not using all of it. The constraint is not the bottleneck.

2

Shadow price > 0 (for a minimisation problem, this means negative)

The constraint is active and binding. The warehouse is hitting its capacity limit in at least one period. If you add one more unit of storage, total cost drops by exactly the shadow price value. The constraint is a genuine bottleneck.

In PuLP you read shadow prices like this:

```
# After solving, read shadow price of a named constraint
m.solve(pulp.PULP_CBC_CMD(msg=0))

for t in range(T):
    pi = m.constraints[f'storage_cap_{t}'].pi
```

What to Watch

Video	What it covers
What is the SHADOW PRICE in Linear Programming? youtube.com/watch?v=4H9COmgZYO8	Clear explanation of shadow prices ,what the value means and how to interpret zero vs nonzero shadow prices.

Deliverable

Write whole code for it.

Then try : tightening the capacity

Change h_{r2}

Change Lead Time

And other factors.

Also observe shadow price.